FINITE RESIDUE FIELD EXTENSION UNDER BOUNDED FIBRES AND WEAK DECENCY

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Residue fields of quasi-finite points. Morphisms, lemma finite-residue-field-extension (cross-chapter, named). The scheme case for U -> X.

Source reference. Original lemma 07U5: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3573–3573. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0ABT: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1421–1421. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

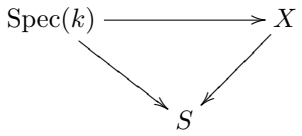
Source reference. Original lemma 047Z: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 3531–3531. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03ZR: focused-seeds/geometry/S9-decent-finite-residue-field-extension-finite/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03IQ: focused-seeds/geometry/S9-decent-finite-residue-field-extension-finite/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let X be a decent algebraic space over S. Consider a commutative diagram*



Assume that the image point $s \in S$ of $\mathrm{Spec}(k) \to S$ is a closed point and that the field extension $k/\kappa(s)$ is finite. Then $\mathrm{Spec}(k) \to X$ is a finite morphism. If $\kappa(s) = k$ then $\mathrm{Spec}(k) \to X$ is a closed immersion.

Proof. By Lemma 07U5 the image point $x \in |X|$ is closed. Let $Z \subset X$ be the reduced closed subspace with $|Z| = \{x\}$ (Properties of Spaces, Lemma 03IQ). Note that Z is a decent algebraic space by Lemma 0ABT. By Lemma 047Z we see that $Z = \mathrm{Spec}(k')$ for some field k'. Of course $k \supset k' \supset \kappa(s)$. Then $\mathrm{Spec}(k) \to Z$ is a finite morphism of schemes and $Z \to X$ is a finite morphism as it is a closed immersion. Hence $\mathrm{Spec}(k) \to X$ is finite (Morphisms of Spaces, Lemma 03ZR). If $k = \kappa(s)$, then $\mathrm{Spec}(k) = Z$ and $\mathrm{Spec}(k) \to X$ is a closed immersion. □